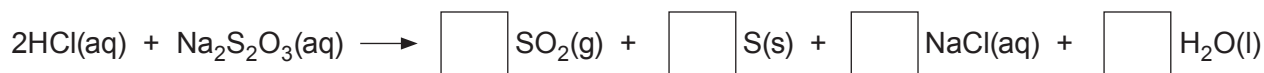


8. (a) A student was investigating how the concentration of sodium thiosulfate solution affects its reaction with hydrochloric acid.

- (i) The reaction taking place can be represented by the following equation, which is **not** balanced.

Write the number **2** in **one of the boxes** on the right hand side in order to balance the equation. [1]



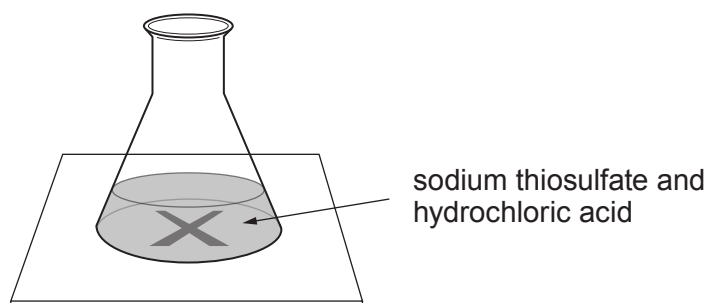
- (ii) During the reaction, the solution becomes cloudy due to the formation of a precipitate. State the meaning of the term *precipitate*. [1]

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(iii) The time taken for the cross to disappear was measured as shown below.



The results are shown in the table.

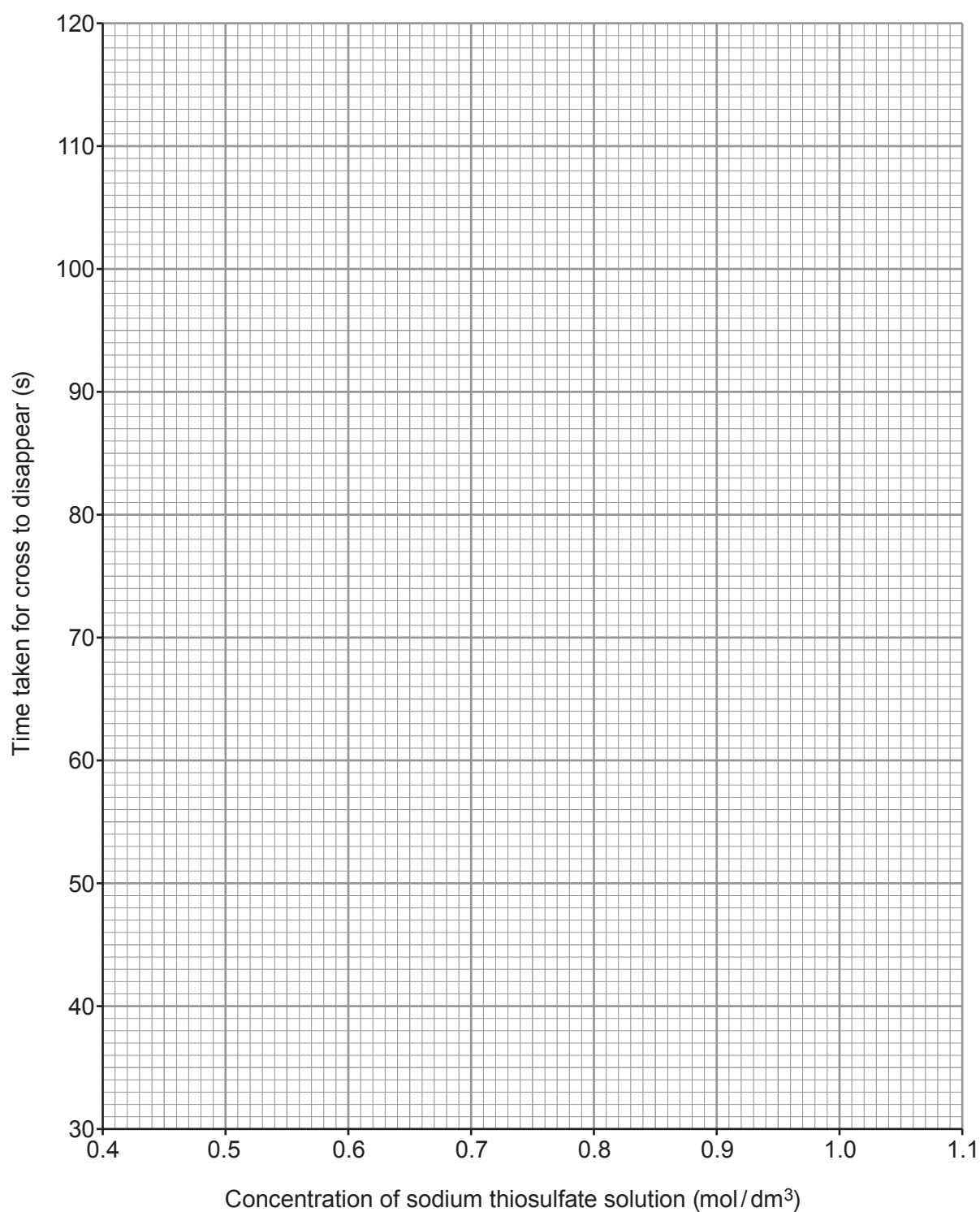
Concentration of sodium thiosulfate solution (mol/dm^3)	Time taken for cross to disappear (s)
1.0	42
0.9	46
0.8	53
0.7	66
0.6	87
0.5	110



Plot the results from the table on the grid. Draw a suitable line.

[3]

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(iv) State how concentration affects the **time taken** for the cross to disappear. [1]

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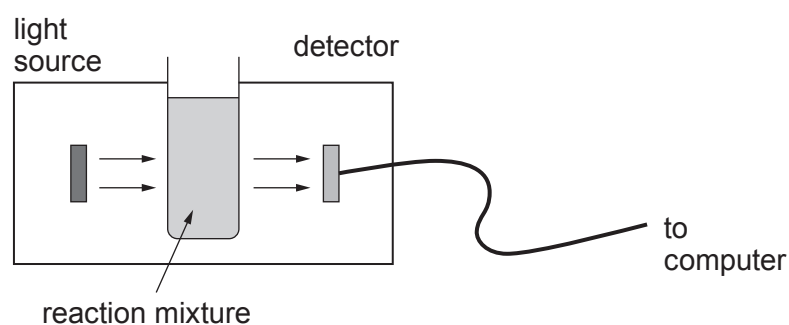
(v) What conclusion can be drawn about the effect of concentration on the **rate** of this reaction? [1]

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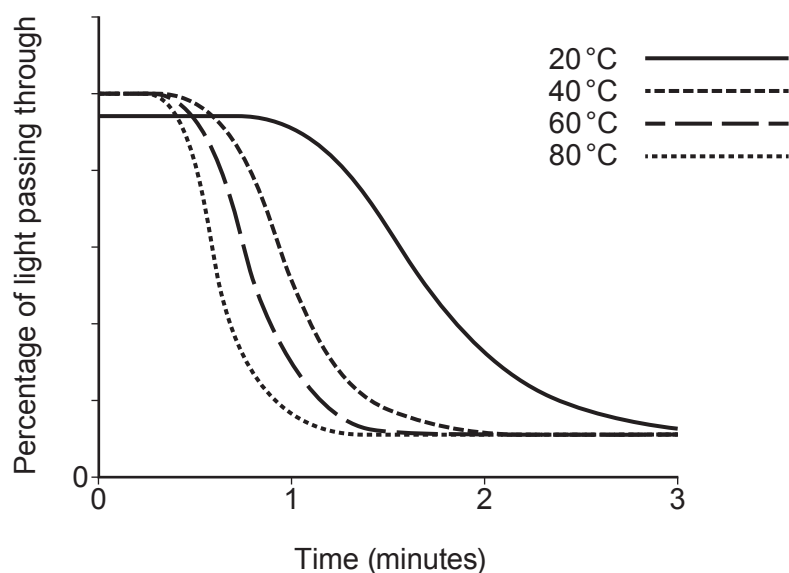
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- (b) A second student used a light sensor to investigate the effect of temperature on the rate of this reaction.



He obtained the following results.



- (i) Give **one** conclusion that can be drawn from these results. State how this is shown by the graph. [2]

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- (ii) Slightly less light passed through the tube at the start of the experiment at 20 °C than in the others. Suggest a possible practical reason for this. [1]

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